

Modelling Assignment

I. Introduction

The task at hand is to design an ontological system for cataloguing library assets for use in an AI-powered search engine for library staff and patrons.

II. Ontology Development Approach

A. Assets to be Catalogued

Library assets are expected to be substantial in number and quite diverse in terms of format and subject matter. Aside from paper-printed books, the library collection is expected to include assets formatted as magazines, pamphlets, and standalone articles, among other formats. Assets are also expected to be presented in multiple types of media, such as print, e-book or electronic file, or HTML-coded for web delivery (Rahmanova, 2025; Wong and Dickson, 2024).

In other words, assets are expected to be describable using multiple overlapping ontological attributes; for example, a book may have several authors of record, belong to several genres, and make use of multiple presentational formats.

B. Challenges Inherent to the Assets

Because library assets will not be uniform or fungible in nature (Rahmanova, 2025; Wong and Dickson, 2024), they will not be easily described by a single class or a single, linear hierarchy of classes and subclasses, leaving aside the likelihood that information about an asset may be missing, incomplete, or inaccurate. And because the number of assets is expected to be substantial, an inherent risk exists that any search engine built for them may return incomplete or irrelevant results,

absent a deliberate effort to design an ontology tightly-scoped for search engine purposes (Huettemann, Mueller and Dinter, 2025).

C. Ontological Design Solution

1. Methodology Followed

The proposed ontology was designed and validated for consistency and coherence using Protégé v. 5.6.7 (Stanford Center for Biomedical Informatics, 2019); Appendices A to C). A visualization of the ontology's class structure was generated by the Protégé Ontograp plugin (protegeproject/ontograp, 2016; Appendix D).

2. Solution Description

Since the library assets at hand are quite diverse, an ontology for them should best describe them using the intersection of numerous classes and subclasses that reflect their significant attributes as relevant to search engine purposes. Restated, a library asset should be described by that cluster of attributes that accurately and adequately assert its uniqueness from any other library asset in a manner amenable to search queries above any other end user purpose (Aman et al, 2023).

Given that approach, the ontology proposed here divides asset attributes between eight top-level classes: audience, descriptive details, author, genre, illustrated, language, publication format, publication type, and publisher (Appendices A and B). Each top-level class is then subdivided, as needed, into subclasses that further define unique itemized entries. For example, the genre class is subclassed between classes for fiction and nonfiction. The fiction subclass is further subclassed between classes for poetry and prose, with the prose subclass subdivided between major prose fiction genres, such as adventure novels, mystery novels, and similar genre books (Appendices A and B).

Critically, to be readily usable in a real world application, the ontological design should primarily focus on end user needs: the ontology should not be designed to fit any particular pre-existing assumptions about its subject matter, but rather should be designed to most accurately and efficiently support end user goals for its practical use (Zhang and Xiao, 2024).

For the task at hand, that means that the proposed ontology revolves around those library asset attributes that users will most likely select to conduct searches, such as the author's name, the asset's genre, and the asset's publication type and format: if a user wants to search for a paperback mystery novel written by a specific author, those same salient attributes need to be at the forefront of the ontology's description of the asset.

Correspondingly, irrelevant attributes are by necessity discarded altogether (Carriero, Groth, and Presutti, 2024); for example, a book's sticker price at a given commercial bookstore on a given date in a given city is irrelevant to a nonprofit community library's search engine, and should be excluded.

3. Solution Advantages

The primary advantages of this approach are precision and modularity or flexibility for future refinement (Lutsyk and Fedasyuk, 2024). This approach is precise in that it can incorporate any clearly-definable and relevant attribute that describes an asset. It's also modular and flexible, in that new attribute classes and subclasses can be added, and deprecated ones removed, as more information about its accuracy and efficiency is gathered from a software implementation's use (Lutsyk and Fedasyuk, 2024).

4. Solution Disadvantages

The proposed ontology also brings with it the distinct disadvantage of increased complexity from the use of numerous classes and subclasses. That increased complexity means that for technical reasons, search performance may be impacted for particularly complicated asset searches, depending on how the ontology is implemented in code (Nasrabadi et al, 2024).

III. Considerations for Practical Application

It should be noted that the proposed ontology may overlap to a significant degree with the core elements of the database schemas to be developed to support the final software product (Hwang et al, 2025; Pierro, 2025). The risk also exists,

however, for significant divergence from the ontology in terms of the structure of the actual code to be built, which could easily result in increased development time and/or heightened complexity of and/or increased maintenance costs for the finished software (Hwang et al, 2025; Pierro, 2025).

Summarized, while the proposed ontology will act as a conceptual map for the code to be developed, it is intended to function as a guidance framework, not as a prescriptive framework that cannot accommodate deviations. It is also expected to change over the course of development as new information comes to light that may be used to refine the ontological class structure for greater accuracy, more efficient schema representations in the finished code, and quicker search engine processing.

IV. Conclusion

Aligned with current industry best practices, the proposed ontology reflects a user-centric focus on its accuracy and efficiency for real world use. While such an approach benefits from increased precision, modularity, and flexibility in defining and describing its subject matter, it also faces the challenge of increased systematic complexity, which may, in turn, increase the complexity and decrease the efficiency of the software code use to implement it.

V. References

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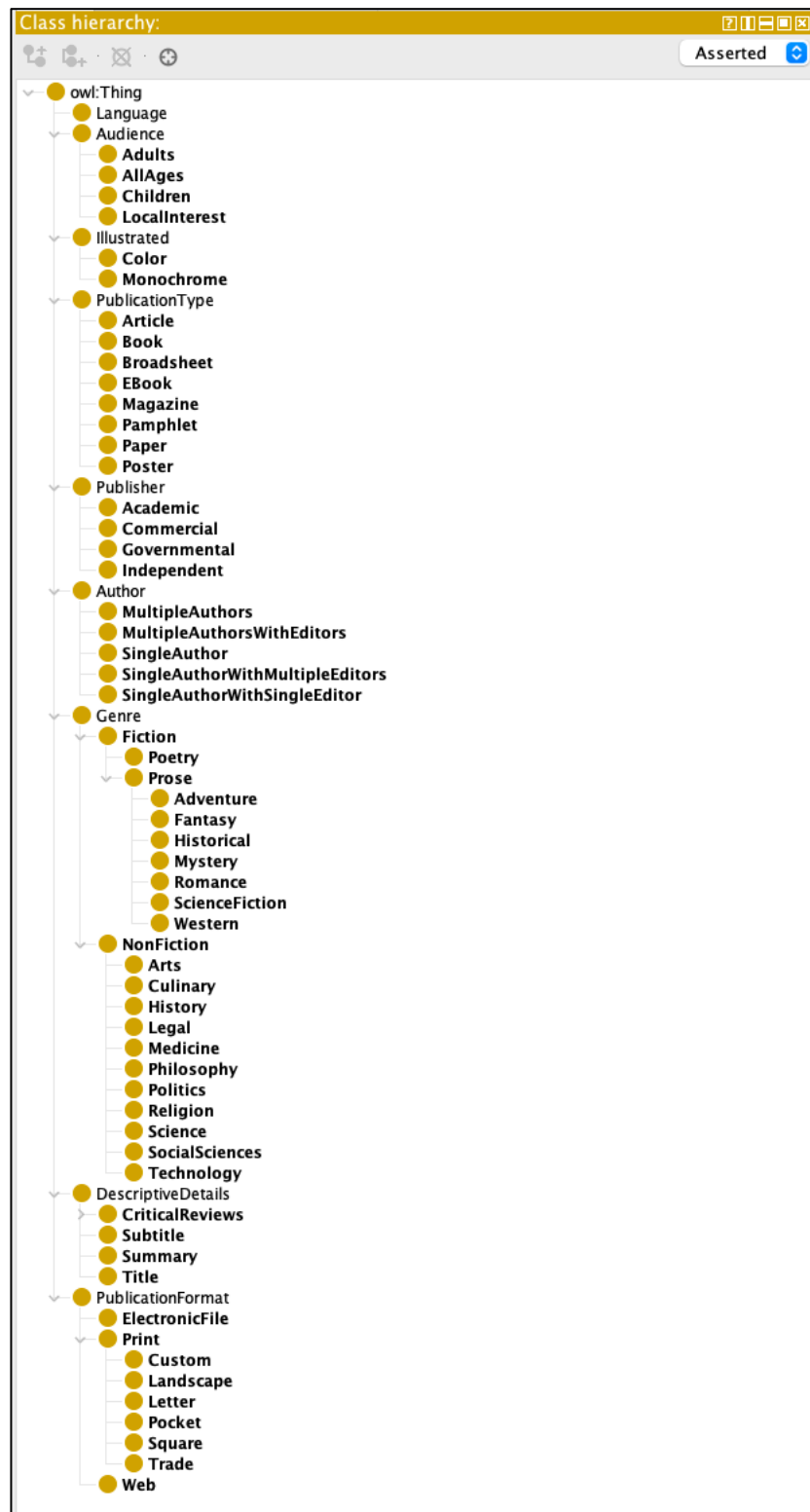
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Appendix A: Ontology Class Structure (Protégé Format)



Appendix B: Ontology Class Structure (CSV / Excel Format, Extract)

	A	B	C
1	Author	Class	
2	SingleAuthor	Class	Author
3	MultipleAuthors	Class	Author
4	SingleAuthorWithSingleEditor	Class	Author
5	SingleAuthorWithMultipleEditors	Class	Author
6	MultipleAuthorsWithEditors	Class	Author
7	Publisher	Class	
8	Commercial	Class	Publisher
9	Academic	Class	Publisher
10	Governmental	Class	Publisher
11	Independent	Class	Publisher
12	PublicationType	Class	
13	Book	Class	PublicationType
14	EBook	Class	PublicationType
15	Magazine	Class	PublicationType
16	Paper	Class	PublicationType
17	Article	Class	PublicationType
18	Pamphlet	Class	PublicationType
19	Broadsheet	Class	PublicationType
20	Poster	Class	PublicationType
21	Genre	Class	
22	Fiction	Class	Genre
23	Prose	Class	Fiction
24	Adventure	Class	Prose
25	Mystery	Class	Prose
26	ScienceFiction	Class	Prose
27	Fantasy	Class	Prose
28	Western	Class	Prose
29	Romance	Class	Prose
30	Historical	Class	Prose

Appendix C: Validation Test of the Ontology Class Structure



Appendix D: Ontograp Visualization of the Ontology Class Structure

